Cloud Computing

Subject: Computer Science | NoteShare Library

CLOUD COMPUTING

1. MODELS
IaaS: Infrastructure (EC2)
PaaS: Platform (Heroku)
SaaS: Software (Gmail)

2. DEPLOYMENT
Public (AWS, Azure, GCP)
Private (on-premise)
Hybrid
Community

3. AWS SERVICES
EC2: Virtual machines
S3: Storage
RDS: Databases
Lambda: Serverless
CloudFront: CDN
Route 53: DNS
IAM: Access control
VPC: Networking

4. VIRTUALIZATION
Hypervisor: Type 1, Type 2
Containers: Docker
Orchestration: Kubernetes

5. SCALING
Vertical: bigger machine
Horizontal: more machines
Auto-scaling
Load balancers

6. STORAGE
Block (EBS)
File (EFS)
Object (S3)

7. DATABASES
Relational (RDS)
NoSQL (DynamoDB)

Data warehouse (Redshift)

8. COST
On-demand
Reserved
Spot instances
Free tier